

Gym Management System

Schema Mapping Document (Final)

SUBSCRIPTION_PLAN

Source CSV File: *subscription_plans.csv*

ERD Attribute	CSV Column	Data Type	Key	Notes
Subscription_Plan	subscription_plan	VARCHAR(50)	PK	Unique plan name; natural key
Monthly_Price	price_per_month	DECIMAL(6,2)		Numeric currency value
Features	features	VARCHAR(255)		Comma-separated list of included features

MEMBER

Source CSV File: *users_data.csv*

ERD Attribute	CSV Column	Data Type	Key	Notes
User_ID	user_id	VARCHAR(20)	PK	Unique member identifier
First_Name	first_name	VARCHAR(50)		
Last_Name	last_name	VARCHAR(50)		
Age	age	INT		Derived/verify against Birthdate if needed
Gender	gender	VARCHAR(20)		
Birthdate	birthdate	DATE		Format: YYYY-MM-DD
Sign_up_Date	sign_up_date	DATE		Format: YYYY-MM-DD
Location	user_location	VARCHAR(50)		Renamed from user_location for ERD consistency
Subscription_Plan	subscription_plan	VARCHAR(50)	FK	References SUBSCRIPTION_PLAN.Subscription_Plan

GYM_BRANCH

Source CSV File: *gym_locations_data.csv*

ERD Attribute	CSV Column	Data Type	Key	Notes
Gym_ID	gym_id	VARCHAR(20)	PK	Unique branch identifier
Location	location	VARCHAR(50)		City name
Gym_Type	gym_type	VARCHAR(20)		e.g., Premium / Budget / Standard

FACILITY

Source CSV File: *gym_locations_data.csv* (derived — 'facilities' column)

Normalization Note (1NF): The original CSV contained Facilities as a comma-separated multi-value column (e.g., "Pool, Sauna, Parking"). This violates 1NF because each cell must hold a single atomic value. To resolve this, the multi-valued attribute was split

into two tables: FACILITY (a lookup table of distinct facility names with a surrogate Facility_ID) and GYM_FACILITY (a junction/bridge table recording one row per gym-facility pair, with a composite PK of Gym_ID + Facility_ID).

ERD Attribute	CSV Column	Data Type	Key	Notes
Facility_ID	— (generated)	INT (auto-increment)	PK	Surrogate key generated during normalization
Facility_Name	facilities (split by comma)	VARCHAR(50)		Distinct value extracted from the comma-separated list

GYM_FACILITY

Source CSV File: *gym_locations_data.csv* (derived — 'facilities' column)

Composite Primary Key: GYM_FACILITY has no surrogate key. Its Primary Key is a composite key formed by (Gym_ID + Facility_ID). This guarantees that each gym-facility combination is unique. Gym_ID is a FK referencing GYM_BRANCH.Gym_ID, and Facility_ID is a FK referencing FACILITY.Facility_ID. Both columns together form the composite PK and must be NOT NULL.

ERD Attribute	CSV Column	Data Type	Key	Notes
Gym_ID	gym_id	VARCHAR(20)	PK / FK	References GYM_BRANCH.Gym_ID
Facility_ID	facilities (split by comma, matched to FACILITY)	INT	PK / FK	One row per (Gym, Facility) pair after splitting

WORKOUT_SESSION

Source CSV File: *checkin_checkout_history_updated.csv*

ERD Attribute	CSV Column	Data Type	Key	Notes
Session_ID	— (generated)	INT (auto-increment)	PK	Surrogate key; no unique ID exists in source file
Gym_ID	gym_id	VARCHAR(20)	FK	References GYM_BRANCH.Gym_ID
User_ID	user_id	VARCHAR(20)	FK	References MEMBER.User_ID
Checkin_Time	checkin_time	DATETIME		Format: YYYY-MM-DD HH:MM:SS
Checkout_Time	checkout_time	DATETIME		Format: YYYY-MM-DD HH:MM:SS; Pending — will be NOT NULL after Stage 3 cleaning
Workout_Type	workout_type	VARCHAR(50)		e.g., Cardio, Yoga, Weightlifting, CrossFit, Swimming, Pilates
Calories_Burned	calories_burned	INT		Pending — will be NOT NULL after Stage 3 cleaning